

$$\frac{d(kp)}{dt} = r(kp - c')$$

$$\Rightarrow k \frac{dp}{dt} = r(kp - c')$$

$$\Rightarrow \frac{dp}{dt} = r \left(p - \frac{c'}{k} \right)$$

Since $k < 1$ $\frac{c'}{k} > c'$ and $\left(p - \frac{c'}{k} \right) < (p - c')$

Since the competitive equilibrium is $\frac{d(p - c')}{dt} = r(p - c')$ which is for $\frac{dc'}{dt} = 0$ reduces to

$$\frac{dp}{dt} = r(p - c')$$

Price is growing more slowly for monopolist and the extraction path is also flatter. (Refer Fig 24.1)

Note: If $\frac{dc'}{dt} \neq 0$, $\frac{dp}{dt} = r \left(p - \frac{c'}{k} \right) + \frac{dc'}{dt} k$

If $\frac{dc'}{dt} < 0$ then $\left| \frac{dc'}{dt} k \right| > \left| \frac{dc'}{dt} \right|$

which implies that monopoly $\frac{dp}{dt}$ is further reduced relative to competitive $\frac{dp}{dt}$

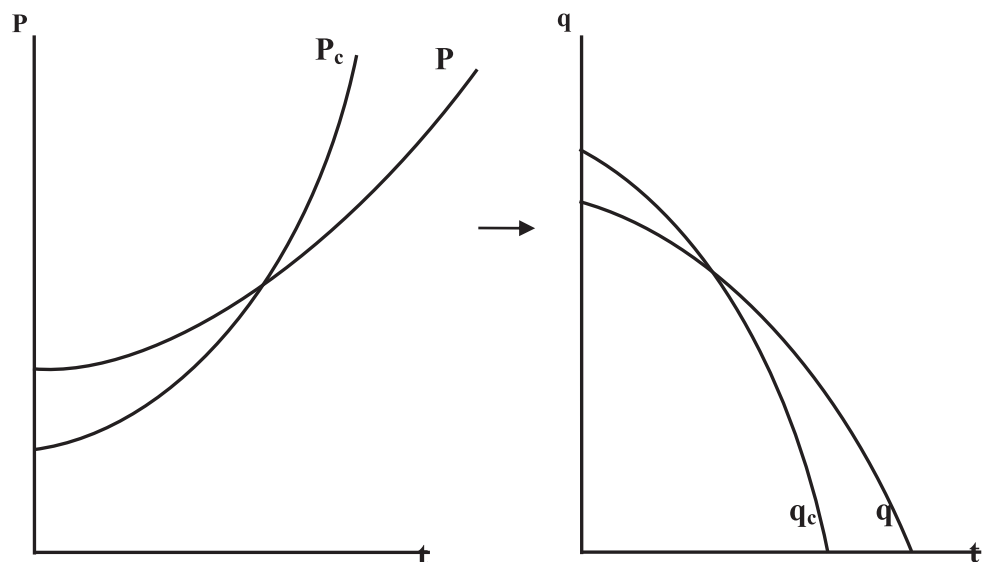


Fig. 24.1: Competitive and Monopoly Price and Extraction Path

24.5 OPTIMAL CONTROL OF A GENERIC RENEWABLE RESOURCE²

Consider the optimal management of a generic renewable resource. Here we have an addition of a biological growth function $g(x_t)$ the problem can be rewritten as

$$\max \int_0^{\infty} [p_t q_t - c(q_t, x_t)] e^{-rt} dt \quad \dots (24.8)$$

$$\text{Subject to } x' = g(x_t) - q_t \quad \dots (24.9)$$

$$\text{where } g(x_t) = \gamma x_t \left(1 - \frac{x_t}{k}\right) \quad \dots (24.10)$$

Here $g(x_t)$ follows the logistic growth law, as used in example of forestry problem, to characterize the behaviour of a renewable natural resource or population, where, γ is the intrinsic growth rate of resource and k is the carrying capacity of environment. As noted earlier, for x_t very small $g(x_t) \cong \gamma x_t$; exponential growth as $x_t \rightarrow k$ $g(x_t) \rightarrow 0$ these relationships are illustrated in Fig.24.2.

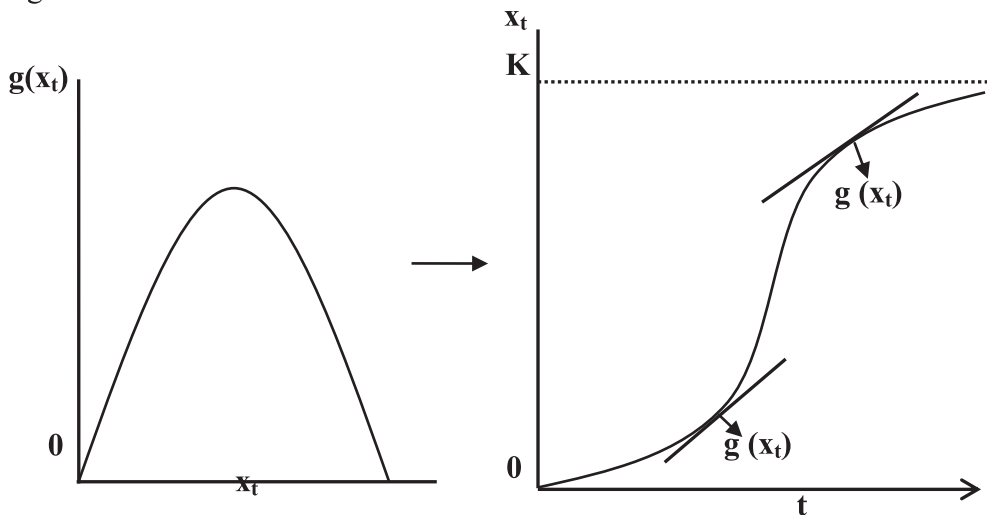


Fig. 24.2: The growth function

The optimal problem can be expressed as

$$H^* = p_t q_t - c(q_t, x_t) + \mu_t (g(x_t) - q_t)$$

So the necessary condition

$$\frac{\partial H^*}{\partial q_t} = 0 \Rightarrow p_t - \frac{\partial c(\cdot)}{\partial q_t} - \mu_t = 0$$

² Fisher Anthony. C, Lecture Notes on Resource and Environmental Economics. Springer Nature, Switzerland AG 2020

Using formula in equation (23.40)

$$\mu' = r\mu_t - \frac{\partial H^*}{\partial x_t} = r\mu_t + \frac{\partial c(.)}{\partial x_t} - \mu_t g'(x_t)$$

$$\Rightarrow \frac{\mu'}{\mu} = r + \frac{\partial c(.)}{\partial x_t} \cdot \frac{1}{\mu} - g'(x_t)$$

This result as in the exhaustible resource problem has an economic interpretation. The new term on right hand side $g'(x)$ provides an extra dividend, the extra growth that results from holding an extra unit of the resource in the stock.

In steady state $\mu' = 0$ so $r = g'(x_t) - \frac{\partial c(.)}{\partial x(t)} \cdot \frac{1}{\mu}$

$$\text{If } \frac{\partial c}{\partial x_t} = 0 \Rightarrow r = g'(x_t)$$

Assuming $r > 0$, this implies that the steady state stock corresponds to $g(x)$ increasing ($g'(x) > 0$) or in other words the stock is to the left of (is less than) the maximum sustainable yield (MSY) stock, as shown in Fig. 24.3.

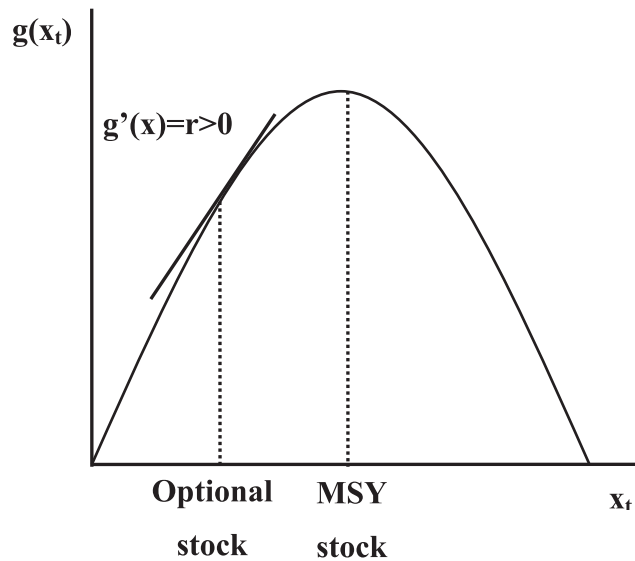


Fig. 24.3: When the discount rate is positive, the optimal stock size is less than MSY stock size.

of course, if $r = 0$ then $g'(x) = 0$ and the MSY stock is optimal as in Fig. 24.4.

we can characterize the MSY stock as follows:

$$g(x_t) = \frac{dx_t}{dt} = \gamma x_t \left(1 - \frac{x_t}{k} \right)$$

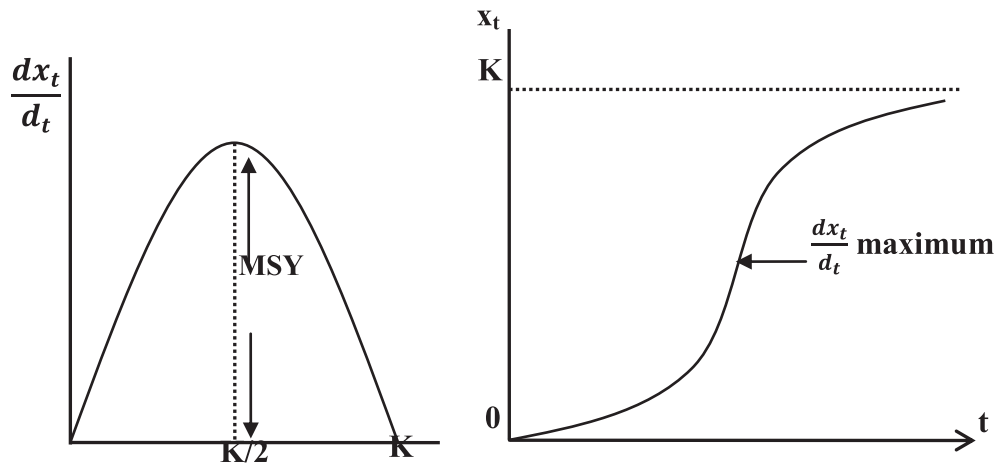


Fig 24.4: Optimal MSY Stock at Discount Rate Zero

$$g'(x_t) = \frac{d}{dt} \frac{dx_t}{dt} = \frac{d}{dt} \left(\gamma x_t - \frac{\gamma x_t^2}{k} \right)$$

Setting $g'(x_t) = 0$

We get $\gamma - 2 \frac{\gamma x_t}{k} = 0$

$$\Rightarrow \frac{2\gamma}{k} x_t = \gamma$$

$$\Rightarrow x_t = k/2$$

Therefore, in absence of discounting the bio-economic optimum is equivalent to what might be called the purely biological optimum, the MSY.

24.6 OPTIMAL DEPLETION OF NON- RENEWABLE RESOURCE

We need to determine the optimal depletion rate for a non-renewable resource (for example oil or coal). Suppose that the output rate now depends on two inputs capital $k(t)$ and resource depletion rate $R(t)$, that is,

$$Y = F(k, R)$$

Now suppose output is either consumed or used to increase the capital stock giving

$$Y = c + \frac{dk}{dt}$$

$$\Rightarrow \frac{dk}{dt} = F(k, R) - c \quad \dots (24.11)$$

where $c = c(t)$ is the consumption rate. If $S(t)$ denotes the resource stock, then we clearly have $\frac{ds}{dt} = -R$...

(24.12)

we wish to maximize the welfare integral

$$W = \int_0^T e^{-\delta t} u(c) dt \quad \dots (24.13)$$

by finding the optimal paths for resource depletion rate, R and consumption rate, c , subject to the constraint (equation 24.11, 24.12) and also with some initial condition for example $k = k_0, s = s_0$ at $t = 0$.

Here the capital k , and resource stock, s are the state variables, the consumption rate, c and depletion rate, R , are control and $F(k, R)$ is the production function.

Because we have two constraints, we need two λ 's, i.e., λ_{1t} and λ_{2t} (two co-state variables)

Formulating Hamiltonian we get,

$$H^* = e^{-\delta t} u(c) + \lambda_{1t} (F(k, R) - c) + \lambda_{2t} (-R)$$

[pl refer to discounted problem in unit 23 for formula]

$$\text{We set } H^* = e^{\delta t} H(.)$$

For necessary condition of optimisation

$$\frac{\partial H^*}{\partial u} = 0 \Rightarrow u'(c) - \mu_{1t} = 0 \Rightarrow \mu_{1t} = u'(c) \quad \dots (24.14)$$

$$\frac{\partial H^*}{\partial R} = 0 \Rightarrow \mu_{1t} \frac{\partial F(.)}{\partial R} - \mu_{2t} = 0 \Rightarrow \mu_{2t} = \mu_{1t} \frac{\partial F(.)}{\partial R} \quad \dots (24.15)$$

Now the co-state equation

$$\mu'_{1t} = \delta \mu_{1t} - \mu_{1t} \frac{\partial F(.)}{\partial k} = \delta \mu_{1t} - \frac{\partial H^*}{\partial k} \quad \dots (24.16)$$

$$\mu'_{2t} = \delta \mu_{2t} - \frac{\partial H^*}{\partial s} = \delta \mu_{2t} \quad \dots (24.17)$$

differentiating both sides of equation (24.15) we get

$$\mu'_{2t} = \mu'_{1t} \frac{\partial F(.)}{\partial R} + \mu_{1t} \frac{d}{dt} \frac{\partial F(.)}{\partial R} \quad \dots (24.18)$$

Using (24.16) in (24.18) we get

$$\begin{aligned} \mu'_{2t} &= \left(\delta \mu_{1t} - \mu_{1t} \frac{\partial F(.)}{\partial k} \right) \frac{\partial F(.)}{\partial R} + \mu_{1t} \frac{d}{dt} \frac{\partial F(.)}{\partial R} \\ &\Rightarrow \delta \mu_{2t} = \delta \mu_{1t} \frac{\partial F}{\partial R} = \delta \mu_{1t} \frac{\partial F(.)}{\partial R} - \mu_{1t} \frac{\partial F(.)}{\partial k} \cdot \frac{\partial F(.)}{\partial R} + \mu_{1t} \frac{d}{dt} \frac{\partial F(.)}{\partial R} \end{aligned}$$

(using 24.17)

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial F(.)}{\partial R} \right) = \frac{\partial F(.)}{\partial k} \frac{\partial F(.)}{\partial R} \quad \dots (24.19)$$

Using equation 24.14 and differentiating that w.r.t 't' and equating with equation (24.16) we get

$$U''(C).c' = \delta(u'(c) - u'(c)) \frac{\partial F(.)}{\partial k} \quad \dots (24.20)$$

An important quantity in economic term is

$$\eta_{(Eta)} = -c \frac{U''(.)}{u'(.)}$$

This relative risk aversion, the inverse of which will give us intertemporal elasticity of substitution EIS More on EIS when we will discuss Ramsey model. In some literature it is called elasticity of marginal utility. It is often assumed constant.

Therefore equation (24.20) can be rewritten as

$$c' = c \left(\frac{\partial F(.)}{\partial k} - \delta \right) / \eta \quad \dots (24.20)$$

Equation (24.19) and (24.20) are basic equation governing the optimal path together with state equations. The equations are non-linear and coupled and to proceed further we must specify the production function let the production function

$$F(.) = k^\alpha R^{1-\alpha} \quad \dots (24.21)$$

where α is positive constant such that $0 < \alpha < 1$

where we find $\frac{\partial F(.)}{\partial k} = k^{\alpha-1} R^{1-\alpha}$, $\frac{\partial F(.)}{\partial R} = (1-\alpha) k^\alpha R^{-\alpha}$

setting $x = k/R$ = capital/resource depletion rate ratio

$$\frac{\partial F(.)}{\partial K} = \alpha X^{\alpha-1} \text{ and } \frac{\partial F(.)}{\partial R} = (1-\alpha) X^\alpha$$

$$\frac{d}{dt} \left(\frac{\partial F(.)}{\partial R} \right) = (1-\alpha) \alpha X^{\alpha-1} x'$$

and equation (24.19) reduces to

$$x' = x^\alpha \quad \dots (24.22)$$

This can be integrated to give

$$x = [(1-\alpha)t + x_0^{1-\alpha}]^{\frac{1}{1-\alpha}} \quad \dots (24.23)$$

where x_0 is constant

We can now deal with (24.20) which gives

$$\frac{c'}{c} = \frac{1}{\eta}(\alpha x^{\alpha-1} - \delta) = \frac{\alpha}{\eta} \frac{x'}{x} - \frac{\delta}{\eta}$$

integrating

$$\log c = \frac{\alpha}{\eta} \log x - \frac{\delta t}{\eta} \log A$$

$$\text{that is } c = Ax^{\alpha/\eta} e^{-\delta t/\eta}$$

and if at time $t = 0$ $c = c_0$ we have

$$c = c_0(x/x_0)^{\alpha/\eta} e^{-\delta t/\eta}$$

$$\text{That is } c = c_0 e^{-\delta t/\eta} [1 + (1-\alpha)t/x_0^{1-\alpha}]^{\frac{\alpha}{\eta(1-\alpha)}} \quad \dots(24.24)$$

We can also determine the other control the depletion Rate R , from

$$\frac{d}{dt} \left(\frac{k}{R} \right) = \frac{Rk' - R'k}{R^2} = \frac{k'}{R} - k \frac{R'}{R^2}$$

Using equation (24.12) we get

$$\text{Given } F = k^\alpha R^{1-\alpha}$$

$$\frac{F}{R} = \left(\frac{k}{R} \right)^\alpha = x^\alpha$$

$$x' = \frac{1}{R}(F - c) - x \frac{R'}{R} = \frac{F}{R} - \frac{c}{R} - x \frac{R'}{R}$$

$$\Rightarrow x' = x^\alpha - \frac{c}{R} - x \frac{R'}{R}$$

Using equation (24.22) in above equation

$$R' = -c/x$$

that is

$$R' = c_0 e^{-\delta t/\eta} [1 + (1-\alpha)t/x_0^{1-\alpha}]^{(\alpha-\eta)/\eta(1-\alpha)} / x_0$$

This can be integrated numerically except in the rather special case when $\alpha = \eta$,

$$R' = -c_0 e^{-\delta t/\eta} / x$$

On integration this becomes

$$R = R_0 - \frac{\eta c_0}{\delta x_0} (e^{-\delta t/\eta} - 1) \quad \dots (24.25)$$

Where the constant of integration has been chosen so that $R = R_0$ at $t = 0$

We can also find k from $x = k/R$ since we know both x and R , finally we can determine S from (24.25) with $\frac{ds}{dt} = -R$

We still have to determine the appropriate values for the constants introduced using boundary condition

$$\begin{aligned} k(0) &= k_0 & k(T) &= k_T \\ s(0) &= s_0 & s(T) &= 0 \end{aligned}$$

A typical solution is shown in Figure (24.5).

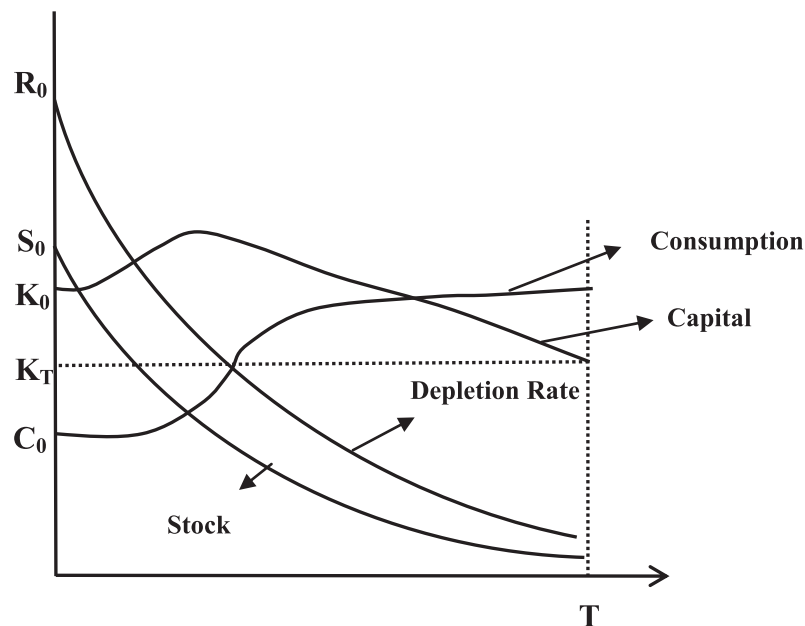


Fig. 24.5: Typical Optimal Trajectories

This presentation closely resembles to the optimal depletion of exhaustible resources by Partha Dasgupta and Geoffrey Heal, 1974. Further models can be referred from Pearce (1975) and Davison (1978).

Check Your Progress 1

- 1) Consider a fishing optimal control problem which is defined by $P'_t = a + bp_t - x_t$ where P_t is fish population at time t . x_t is fishing intensity or catch (a & b are constants). If r is the discount rate and objective function

$$V(.) = \int_0^{\infty} e^{-rt} u(x_t) dt, u(x_t) \text{ is utility function of consumption i.e., } c_t = x_t$$

- State the transversality condition
- Find the optimal consumption, x_t if $u(x_t) = L_n x_t$

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24.7 NEOCLASSICAL INVESTMENT THEORY

Consider a production process with single output. Let the production function we assume that the production function has following properties

- i) $F(k,0) = 0 \forall k$ & $F(0,L) = 0 \forall L$ so that both factors are essential.
- ii) $\frac{\partial F(.)}{\partial k} > 0$ and $\frac{\partial F(.)}{\partial L} > 0$ so more of either factor implies more output.
- iii) $\frac{\partial^2 F(.)}{\partial k^2} \times \frac{\partial^2 F(.)}{\partial L^2} - \left(\frac{\partial^2 F(.)}{\partial k \partial L} \right) < 0$ so that the production function is concave.

We often also assume constant returns to scale so that $F(\lambda k, \lambda L) = \lambda F(k, L) \forall \lambda > 0$. Consider profit, $R(t)$ is defined by

$$R(t) = p(t) F(k_t, L_t) - w(t) L(t) - g(t) I(t) \quad \dots (24.26)$$

where

$F(k_t, L_t)$ = level of output

$p(t)$ = price of output

$w(t)$ = wage

$L(t)$ = labor

$q(t)$ = price of unit investment

$I(t)$ = Investment

Assumption: firm takes the price (p_t, q_t, w_t) which prevail at each 't'. Investment I_t don't affect the prices w_t and q_t for each time t.

We define present value as the integral of discounted net profit with rate of discount is a constant without loss of generality

$$w = \int_0^{\infty} e^{-rt} R(t) dt \quad \dots (24.27)$$

Present value is maximized subject to constraints. First the rate of change of the flow of capital services is proportional to the flow of net investment. Net investment is equal to total investment less replacement, where replacement is proportional to capital stock

$$\frac{dk}{dt} = I(t) - \delta k(t) \quad \dots (24.28)$$

We can say investment is rate of change in capital plus depreciation.

where $\frac{dk}{dt}$ is the time rate change of flow of capital services at time t. we will use

k in place of $k(t)$, I in place of $I(t)$ and so on.

Therefore we can write Hamiltonian (please refer to the discounted problem in unit 23)

$$H^* = PF(k, L) - wL - qI + \mu_t(I - \delta k) \quad \dots (24.29)$$

The necessary conditions for optimisation (note here we have two control variables: L & I and one state variable) are:

$$\frac{\partial H^*}{\partial L} = 0 \Rightarrow p \frac{\partial F(.)}{\partial L} - w \Rightarrow \frac{\partial F}{\partial L} = \frac{w}{p} \quad \dots (24.30)$$

$$\frac{\partial H^*}{\partial I} = 0 \Rightarrow -q + \mu_t = 0 \Rightarrow \mu_t = q \quad \dots (24.31)$$

equation (24.30) is the equation for dynamic marginal productivity of Labor

From co-state equation we get

$$\mu'_t = r\mu_t - p \frac{\partial F(.)}{\partial k} + \mu_t \delta \quad \dots (24.32)$$

Using equation (24.31) in equation (24.32) we get

$$\begin{aligned} q' &= (r + \delta)\mu_t - p \frac{\partial F(.)}{\partial k} \\ \Rightarrow \frac{\partial F(.)}{\partial k} &= \frac{q(r + \delta) - q'}{p} = \frac{c}{p} \quad \dots (24.33) \end{aligned}$$

where $c = q(r + \delta) - q'$

equation (24.33) defines the implicit rental value of capital services supplied by the firm itself.

Once we know the production function we can calculate marginal productivity of labour and capital. Because r and δ for the model are taken constant, equation (24.33) is a 1st order differential equation that needs to be solved in order to find rental to capital.

24.8 OPTIMAL GROWTH MODEL

This model is also known as the Ramsey Model. For sake of simplicity consider a firm which is a representative agent for economy. The firm produces as well consume as a household. The firm maximizes its profit that it maximizes the difference between its income (output) and the cost of production. Let

$$y(t) = \text{output} = F(k_t, L_t)$$

$$w(t) = \text{Unitary cost of hiring a Labor}$$

$$r(t) = \text{Unitary cost of borrowing capital}$$

$$\delta = \text{rate of depreciation for capital} = \text{a constant } \delta \in (0, 1)$$

whatever the firm produces the household consumes that and the balance production after consumption is invested back in firm. The household buys the product. The household has two sources of income i.e., wage from firm and interest on capital. Household buys at price p_t per unit output. Without loss of